Python Programming Foundations

Week 7 – RBA & Project Framing

Day 3: "Approval Through Clear Scope & Structure"

1

Approval Is About Fit

Approval is not about whether an idea sounds exciting.

Approval asks whether the project fits

- the course,
- the time available,
- the developer's current skills,
- the need for clear explanation.

2

Approval Is About Fit

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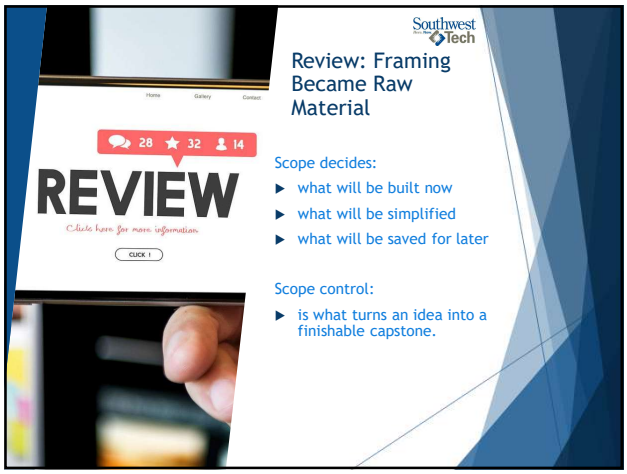
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Review: Framing Became Raw Material
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REVIEW
Click here for more information
CLICK !
Scope decides:
what will be built now
what will be simplified
what will be saved for later
Scope control:
is what turns an idea into a finishable capstone.

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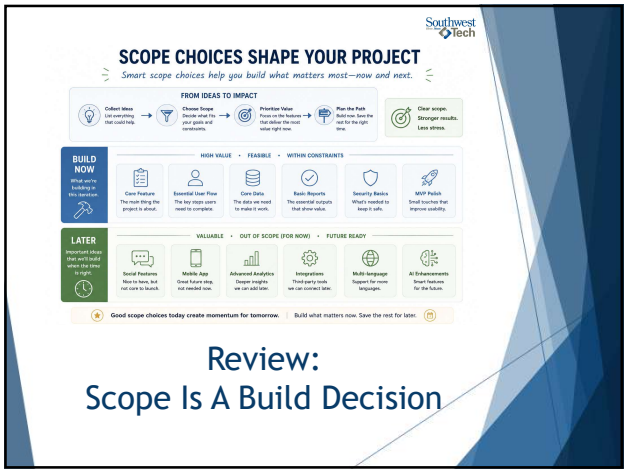
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SCOPE CHOICES SHAPE YOUR PROJECT
Smart scope choices help you build what matters most—now and next.
FROM IDEAS TO IMPACT
Clear scope. Stronger results. Less stress.
BUILD NOW
HIGH VALUE • FEASIBLE • WITHIN CONSTRAINTS
Core Feature
Essential User Flow
Core Data
Basic Reports
Security Basics
MVP Polish
LATER
VALUABLE • OUT OF SCOPE (FOR NOW) • FUTURE READY
Social Features
Mobile App
Advanced Analytics
Integrations
Multi-language
AI Enhancements
Good scope choices today create momentum for tomorrow.
Build what matters now. Save the rest for later.
Review:
Scope Is A Build Decision

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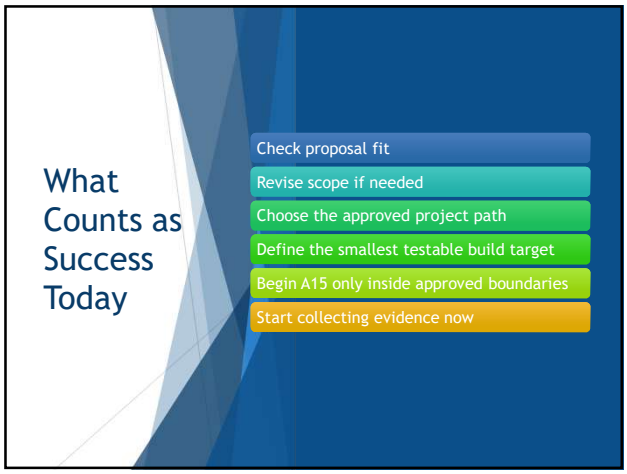
What Counts as Success Today
Check proposal fit
Revise scope if needed
Choose the approved project path
Define the smallest testable build target
Begin A15 only inside approved boundaries
Start collecting evidence now

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CAPSTONE LAUNCH SUCCESS PATTERN
A clear pattern helps you launch with confidence and stay on track.
1 CHECK PROPOSAL FIT
2 REVISE SCOPE IF NEEDED
3 CHOOSE APPROVED PATH
4 DEFINE FIRST BUILD TARGET
5 STAY INSIDE APPROVED BOUNDARIES
6 START COLLECTING EVIDENCE
Follow the pattern. Build with focus. Finish with confidence.
Small steps. Clear choices. Strong results.
Stay Focused
Follow the Plan
Show Your Work
Create Impact
Today's Success

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What We Will Use Today
approval criteria
project path comparison
first build target
run instructions
validation evidence
AI accountability

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TODAY'S LAUNCH TOOLS
Use these tools to launch your capstone with clarity and confidence.
1. APPROVAL CRITERIA
2. PROJECT PATH
3. FIRST BUILD TARGET
4. RUN INSTRUCTIONS
5. VALIDATION EVIDENCE
6. AI ACCOUNTABILITY
Toolbox

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What We Will “Skip” For Now

- final polish
- presentation rehearsal
- optional features
- unapproved expansions
- professional packaging

Today we approve the scope and start the smallest useful build.

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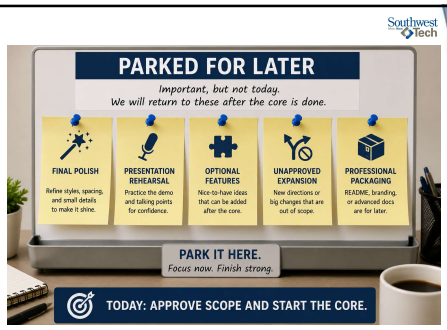

Parked For Later

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Project Approval/ Revision

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Approval Criteria

An approvable proposal shows:

- clear purpose
- realistic features
- expected inputs and outputs
- likely structure
- risks or unknowns
- AI-use plan
- why the scope is realistic

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Approval Criteria

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Project Path Changes Complexity

Possible capstone paths:

- console utility
- data-driven tool
- file-based tracker
- API-style fetcher
- light architecture-preview project, if approved

Each path carries different risks.

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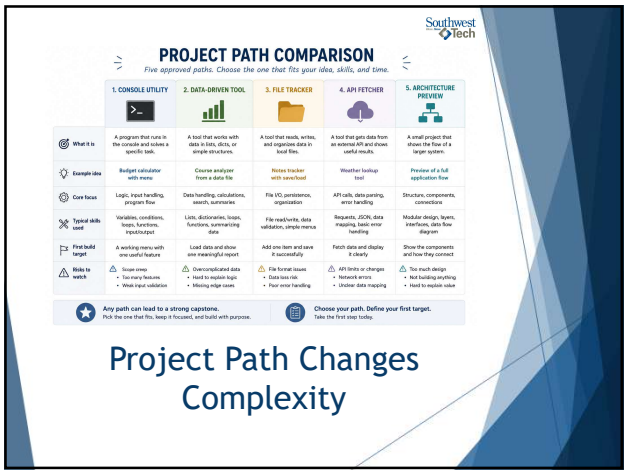
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PROJECT PATH COMPARISON
Five approved paths. Choose the one that fits your idea, skills, and time.
1. CONSOLE UTILITY
2. DATA-DRIVEN TOOL
3. FILE TRACKER
4. API FETCHER
5. ARCHITECTURE PREVIEW
What it is
Example idea
Budget calculator with menu
Course analyzer from a data file
Notes tracker with save/load
Weather lookup tool
Preview of a full application flow
Core focus
Logic, input handling, program flow
Data handling, calculations, search, summaries
File I/O, persistence, organization
API calls, data parsing, error handling
Structure, components, connections
Typical skills used
Variables, conditions, loops, functions, input/output
Lists, dictionaries, loops, functions, summarizing data
File read/write, data validation, simple menus
Requests, JSON, data mapping, basic error handling
Modular design, layers, interfaces, data flow diagram
First build target
A working menu with one useful feature
Load data and show one meaningful report
Add one item and save it successfully
Fetch data and display it clearly
Show the components and how they connect
Risks to watch
Scope creep
Too many features
Weak input validation
Overcomplicated data
Hard to explain logic
Missing edge cases
File format issues
Data loss risk
Poor error handling
API limits or changes
Network errors
Unclear data mapping
Too much design
Not building anything
Hard to explain value
Any path can lead to a strong capstone.
Pick the one that fits, keep it focused, and build with purpose.
Choose your path. Define your first target.
Take the first step today.
Project Path Changes Complexity

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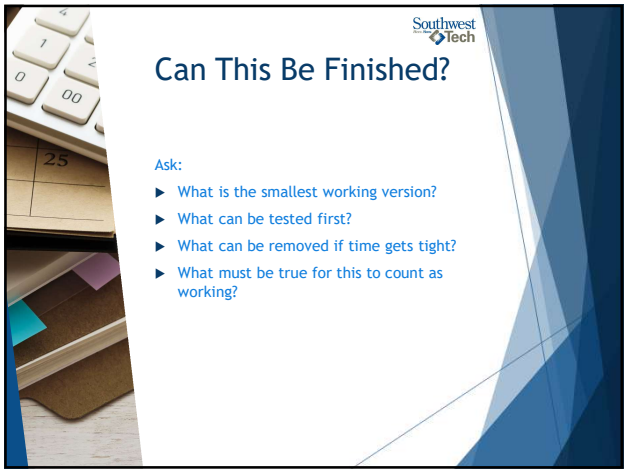
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Can This Be Finished?
Ask:
What is the smallest working version?
What can be tested first?
What can be removed if time gets tight?
What must be true for this to count as working?

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FINISHABILITY: BUILD WHAT YOU CAN FINISH
A clear boundary today helps you deliver a working project.
SMALL, TESTABLE CORE
Build this now.
Core logic that solves the main problem
Basic input or simple menu
Clear output or result
Works end to end
Goal: Prove the idea works with a simple, complete experience.
LATER FEATURES
Save these for after the core works.
Advanced options and settings
Reports and visualizations
Cloud sync or external integrations
Polish, themes, and extra content
Goal: Add value and polish after the core is stable and testable.
CHECK YOUR PLAN
1. SMALLEST WORKING VERSION?
What is the smallest thing I can build that still proves the idea works?
2. TEST FIRST?
Can I test this core early to confirm it works before adding more?
3. REMOVE IF TIGHT?
What features can I remove or delay if time or scope gets tight?
4. COUNTS AS WORKING?
Does the core deliver real value and solve the main problem for the user?
You don't have to build everything.
Finish a focused core. Then make it even better.
Finishability is a feature.
A finished core is better than a perfect plan.
Can This Be Finished?

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Can This Be Explained?

Ask:

- Can I describe the main logic?
- Can I explain the data?
- Can I show validation evidence?
- Can I explain AI contributions, if any?

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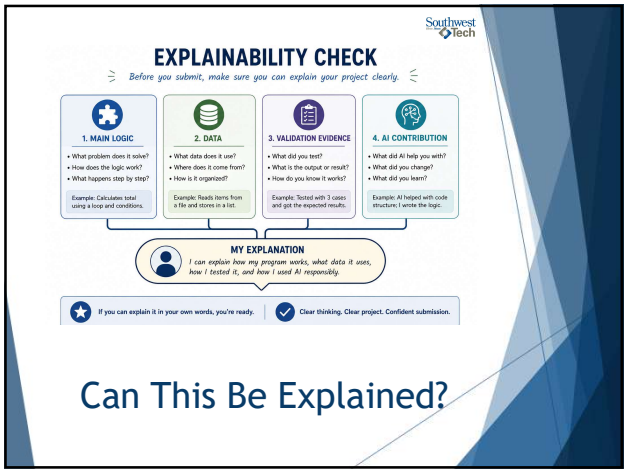

Can This Be Explained?

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Revision Is Not Rejection

If the proposal needs revision, that is not failure.

Targeted revision protects the student from building something too large, unclear, or hard to explain.

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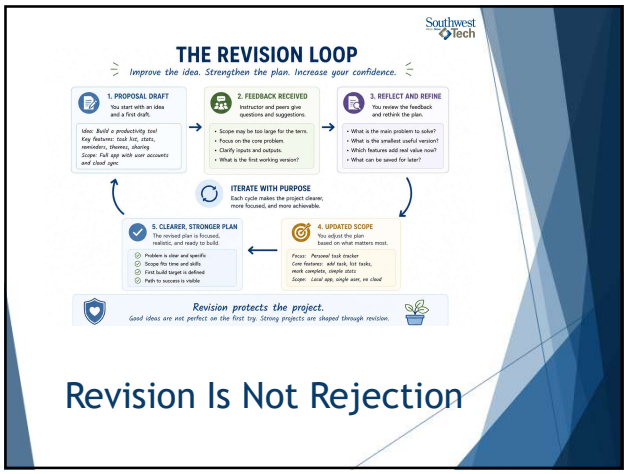
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THE REVISION LOOP
Improve the idea. Strengthen the plan. Increase your confidence.
1. PROPOSAL DRAFT
You start with an idea and a first draft.
Idea: Build a productivity tool
Key features: task list, stats, reminders, themes, sharing
Scope: Full app with user accounts and cloud sync
2. FEEDBACK RECEIVED
Instructor and peers give questions and suggestions.
• Scope may be too large for the term.
• Focus on the core problem.
• Clarify inputs and outputs.
• What is the first working version?
3. REFLECT AND REFINE
You review the feedback and rethink the plan.
• What is the main problem to solve?
• What is the smallest useful version?
• Which features add real value now?
• What can be saved for later?
ITERATE WITH PURPOSE
Each cycle makes the project clearer, more focused, and more achievable.
4. UPDATED SCOPE
You adjust the plan based on what matters most.
Focus: Personal task tracker
Core features: add task, list tasks, mark complete, simple stats
Scope: Local app, single user, no cloud
5. CLEARER, STRONGER PLAN
The revised plan is focused, realistic, and ready to build.
Problem is clear and specific
Scope fits time and skills
First build target is defined
Path to success is visible
Revision protects the project.
Good ideas are not perfect on the first try. Strong projects are shaped through revision.
Revision Is Not Rejection

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Demo #1
"Capstone Proposal Example"

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What to Watch For
clear purpose
focused feature list
structure choice
risk notes
AI-use boundary
realistic scope
Q2
Q3

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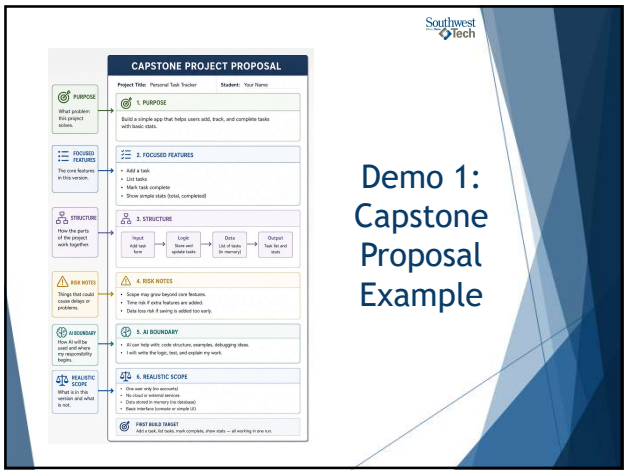
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CAPSTONE PROJECT PROPOSAL
Project Title: Personal Task Tracker
Student: Your Name
PURPOSE
What problem this project solves.
1. PURPOSE
Build a simple app that helps users add, track, and complete tasks with basic stats.
FOCUSED FEATURES
The core features in this version.
2. FOCUSED FEATURES
Add a task
List tasks
Mark task complete
Show simple stats (total, completed)
STRUCTURE
How the parts of the project work together.
3. STRUCTURE
Input
Add task form
Logic
Store and update tasks
Data
List of tasks (in memory)
Output
Task list and stats
RISK NOTES
Things that could cause delays or problems.
4. RISK NOTES
Scope may grow beyond core features.
Time risk if extra features are added.
Data loss risk if saving is added too early.
AI BOUNDARY
How AI will be used and where my responsibility begins.
5. AI BOUNDARY
AI can help with: code structure, examples, debugging ideas.
I will: write the logic, test, and explain my work.
REALISTIC SCOPE
What is in this version and what is not.
6. REALISTIC SCOPE
One user only (no accounts)
No cloud or external services
Data stored in memory (no database)
Basic interface (console or simple UI)
FIRST BUILD TARGET
Add a task, list tasks, mark complete, show stats — all working in one run.
Demo 1: Capstone Proposal Example

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Demo #2
“Structure Options & Approval Fit”

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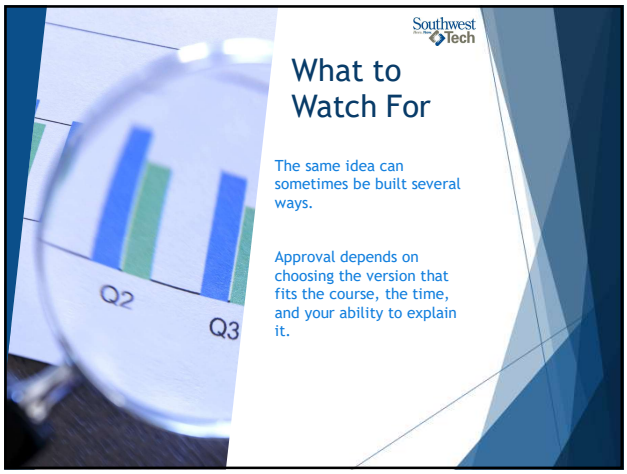
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What to Watch For
The same idea can sometimes be built several ways.
Approval depends on choosing the version that fits the course, the time, and your ability to explain it.
Q2
Q3

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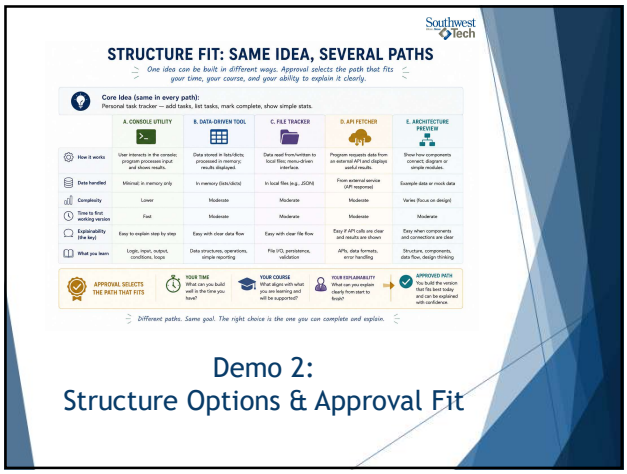
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STRUCTURE FIT: SAME IDEA, SEVERAL PATHS
One idea can be built in different ways. Approval selects the path that fits your time, your course, and your ability to explain it clearly.
Core idea (same in every path):
Personal task tracker — add tasks, list tasks, mark complete, show simple stats.
A. CONSOLE UTILITY
B. DATA-DRIVEN TOOL
C. FILE TRACKER
D. API FETCHER
E. ARCHITECTURE PREVIEW
How it works
Data handled
Complexity
Time to first working version
Explainability (the key)
What you learn
APPROVAL SELECTS THE PATH THAT FITS
YOUR TIME
YOUR COURSE
YOUR EXPLAINABILITY
APPROVED PATH
Different paths. Same goal. The right choice is the one you can complete and explain.
Demo 2:
Structure Options & Approval Fit

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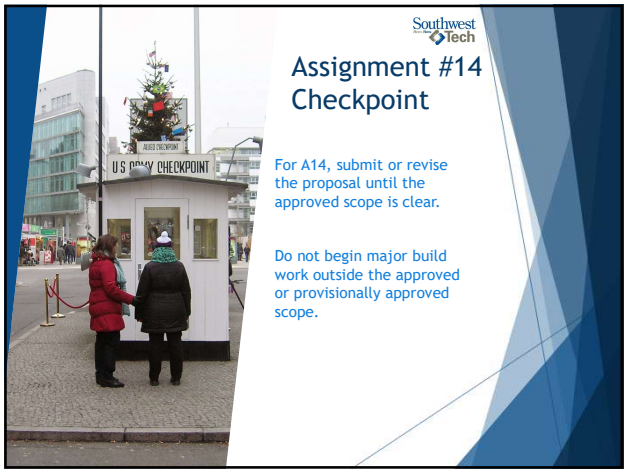
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U.S. DAILY CHECKPOINT
Assignment #14 Checkpoint
For A14, submit or revise the proposal until the approved scope is clear.
Do not begin major build work outside the approved or provisionally approved scope.

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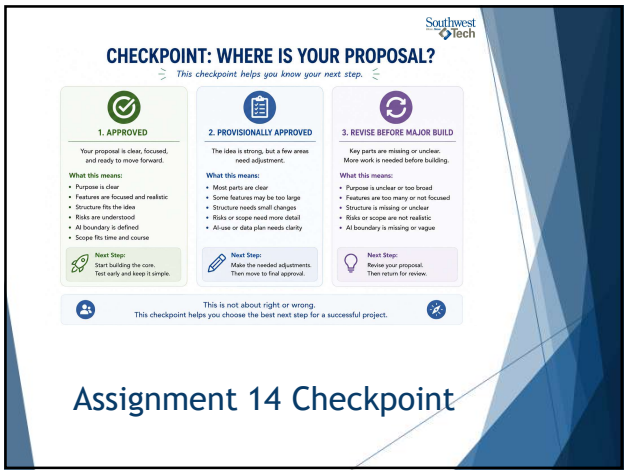
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CHECKPOINT: WHERE IS YOUR PROPOSAL?
This checkpoint helps you know your next step.
1. APPROVED
Your proposal is clear, focused, and ready to move forward.
What this means:
Purpose is clear
Features are focused and realistic
Structure fits the idea
Risks are understood
AI boundary is defined
Scope fits time and course
Next Step:
Start building the core.
Test early and keep it simple.
2. PROVISIONALLY APPROVED
The idea is strong, but a few areas need adjustment.
What this means:
Most parts are clear
Some features may be too large
Structure needs small changes
Risks or scope need more detail
AI-use or data plan needs clarity
Next Step:
Make the needed adjustments.
Then move to final approval.
3. REVISE BEFORE MAJOR BUILD
Key parts are missing or unclear.
More work is needed before building.
What this means:
Purpose is unclear or too broad
Features are too many or not focused
Structure is missing or unclear
Risks or scope are not realistic
AI boundary is missing or vague
Next Step:
Revise your proposal.
Then return for review.
This is not about right or wrong.
This checkpoint helps you choose the best next step for a successful project.
Assignment 14 Checkpoint

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Assignment #15

Once your project is approved or provisionally approved, begin Assignment 15.

Start with the smallest testable core, not the full imagined project

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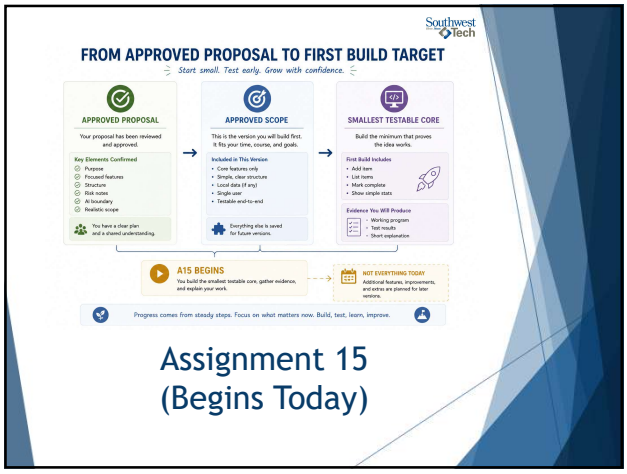

Assignment 15 (Begins Today)

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First Build Target

Your first build target should answer:

- What file or function starts the project?
- What input can I test?
- What output proves progress?
- What will I validate first?

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FIRST-BUILD PLANNING
A few clear answers today make your first build simple and successful.
Approved Scope
First Build Target
Smallest testable core
A15 Begins
Build, test, and learn
PLAN YOUR FIRST STEP
1. WHAT FILE OR FUNCTION STARTS?
Choose the first piece of code you will create.
Example:
main.py
or add_task() function
2. WHAT INPUT CAN I TEST?
Pick a simple, realistic input you can try right away.
Example:
A task name typed in
"Buy groceries"
3. WHAT OUTPUT PROVES PROGRESS?
Decide what you will see that shows it is working.
Example:
Task appears in the list
Total tasks = 1
4. WHAT VALIDATION FIRST?
Choose one basic check to keep things reliable.
Example:
Task name is not empty
No duplicates allowed
KEEP IT SMALL
One simple step. One clear test. One visible result. That is progress.
YOU CAN GROW IT LATER
Extra features, improvements, and polish come after the core works.
First Build Target

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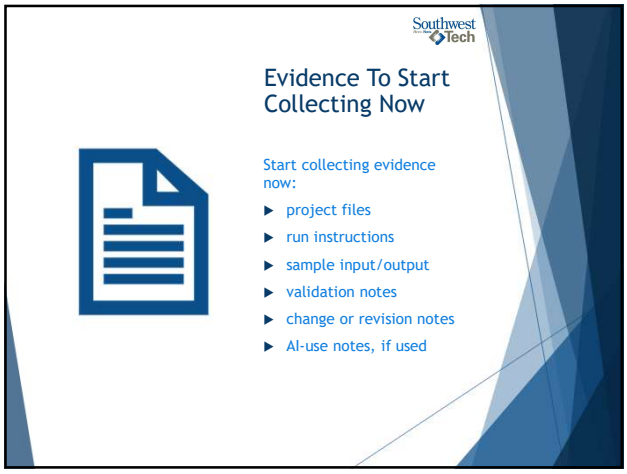
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Evidence To Start Collecting Now
Start collecting evidence now:
project files
run instructions
sample input/output
validation notes
change or revision notes
AI-use notes, if used

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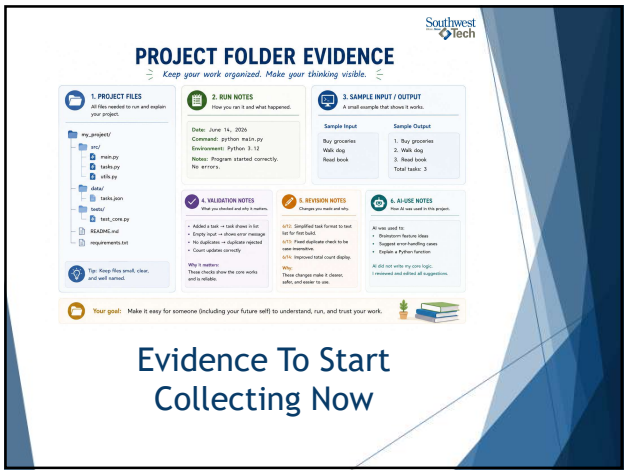
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PROJECT FOLDER EVIDENCE
Keep your work organized. Make your thinking visible.
1. PROJECT FILES
2. RUN NOTES
3. SAMPLE INPUT / OUTPUT
4. VALIDATION NOTES
5. REVISION NOTES
6. AI-USE NOTES
Your goal: Make it easy for someone (including your future self) to understand, run, and trust your work.
Evidence To Start Collecting Now

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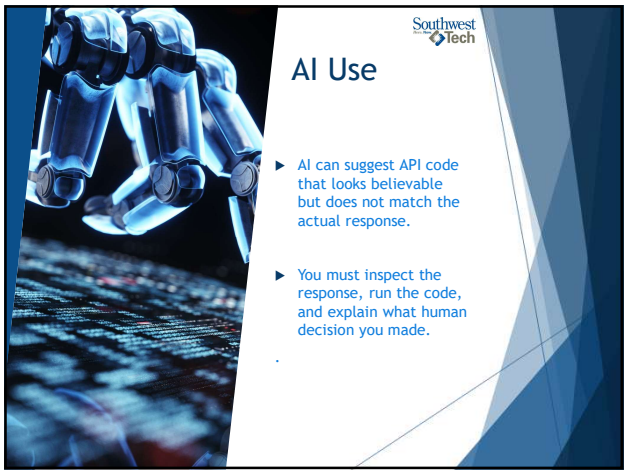

AI Use

- AI can suggest API code that looks believable but does not match the actual response.
- You must inspect the response, run the code, and explain what human decision you made.

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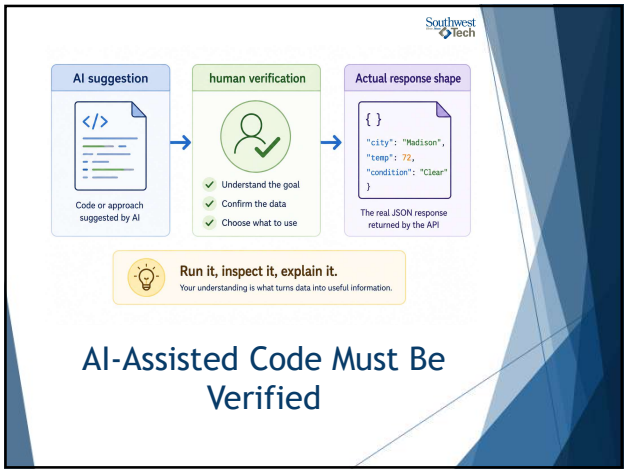

AI-Assisted Code Must Be Verified

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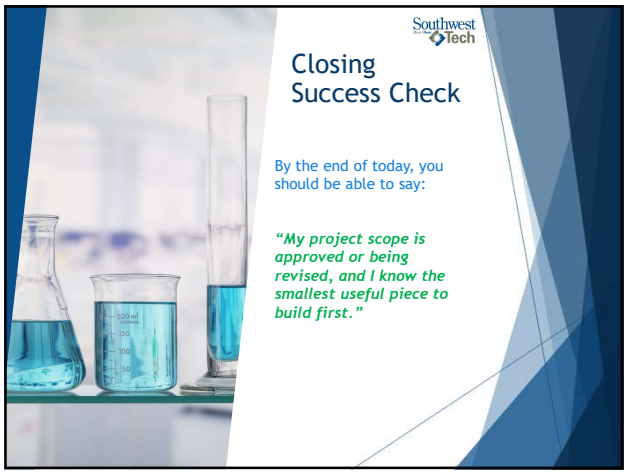

Closing Success Check

By the end of today, you should be able to say:

"My project scope is approved or being revised, and I know the smallest useful piece to build first."

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Success Check

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